Received 4 May 2024
Revised 14 October 2024
7 December 2024
Accepted 13 January 2025

The moderating role of perceived trust in predicting the adoption intention of consumers toward online pharmacies using the application of UTAUT model

Sanjeev Saraswat
Institute of Business Management, GLA University, Mathura, India, and

Pankaj Singh
School of Business, Galgotias University, Greater Noida, India

Abstract

Purpose – This study aims to identify the factors influencing consumer perception and adoption intention toward online pharmacy.

Design/methodology/approach – A questionnaire was formulated based on the unified theory of acceptance and use of technology (UTAUT) framework scales available in the existing literature. An aggregate of 451 respondents participated in the survey. To test the formulated hypothesis, confirmatory factor analysis and structure equation modeling are used for analyzing the structural relationships between dependent and independent variables.

Findings – This analysis inspects the significant effect of perceived trust, consumer perception and different UTAUT framework constructs on the adoption intention of online pharmacy consumers. The results determine the substantial moderating effect of perceived trust between adoption intention and consumers' perception of online pharmacies.

Research limitations/implications – This research extended the application of the UTAUT model with the adoption intention of online pharmacy consumers. Only specific constructs have been selected from the UTAUT model in this study.

Practical implications – The study outcomes are useful to pharmacy managers to realize the role of technological platforms in facilitating the online selling of medicines to pharmacy consumers.

Social implications – The findings are helpful for pharmaceutical companies to improve their existing online buying process for pharmacy consumers seeking purchase of medicines through online pharmacies.

Originality/value – This analysis is unique and contains newness in the form of the moderating role of perceived trust between constructs of UTAUT and customer adoption of online pharmacies.

Keywords Online pharmacies, Adoption intention, Consumer perception, Perceived trust, UTAUT, SEM

Paper type Research paper

1. Introduction

Over the years, the internet has evolved as a core platform of communication infrastructure among corporations and consumers worldwide (Wirtz *et al.*, 2010). With the consistent hike in access to the internet worldwide, rising users of the internet and smartphones, electronic communication is estimated to observe a 19.3% compound annual growth rate (CAGR) till

International Journal of Pharmaceutical and Healthcare Marketing
Vol. 19 No. 4, 2025
pp. 964-980
© Emerald Publishing Limited
1750-6123
DOI 10.1108/IJPHM-05-2024-0042

The authors extend sincere thanks to the Professor Avinandan Mukherjee Editor, International Journal of Pharmaceutical and Healthcare Marketing and anonymous reviewers for their insightful criticism and helpful suggestions, which have greatly raised the caliber of this work.

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

2026 (Teodorescu, *et al.*, 2023). The increasing access to internet facilities opens the door for plenty of opportunities for various service sectors and marketers to provide their offerings to the masses virtually (Wen and Tan, 2003). Many service sectors enhance their customer base by concentrating on online customers including retail, education, tourism, hospitality and health care (Altinay and Poudel, 2015). In case of the health-care sector, pharmacy services are considered an integral, inseparable and indispensable part of medical amenities that are offered to customers (Worlu *et al.*, 2016). A significant volume of clinical service providers have transformed their services to virtual platforms using these tools and simultaneously intermediating between patients and pharmacies wherever needed (Dubey and Verma, 2022).

The online pharmacy service extends society's significant relief especially during the COVID-19 pandemic as increasing waiting lines of patients in hospitals and social distancing as the core adaptive strategy of dealing with the COVID-19 pandemic globally (Elbeddini and Yeats, 2020). Significant transformations in existing clinical and strategic policies are needed to tackle the impact of pandemics and online pharmacy can be one such strategic push (Tan *et al.*, 2020). Online pharmacies are internet-based platforms that allow the buying and selling of pharmaceutical drugs to patients and deliver them using an e-commerce supply-chain tool (Mendoza, 2014). However, online pharmacies use e-mail, telecommunication and video calling options to support the health and pharmaceutical sectors. Subsequently, the online pharmacy industry includes web-based platforms that provide virtual access to medicines to its users rather than physically contacting the retail service providers (Srivastava and Raina, 2021).

The rising and unfulfilled demands of the booming Asia Pacific region in the healthcare sector make it an enticing opportunities destination as far as the global online pharmacy industry is concerned (Connell, 2008). The Indian e-pharmacy market is estimated to reach a US$3,657 valuation by 2022, reporting a CAGR of 63% (Dcruz *et al.*, 2022). The intrusion of COVID-19 in Indian territories in the first quarter of 2020 pushed the Indian Government to go for lockdown, resulting in the double-engine growth of online pharmacies within the beginning phase of the lockdown's introduction. The market share of online pharmacies in India expanded twofold from its prepandemic coverage of over three million to even more than six million users in the initial part of the COVID-19 pandemic break-down (Tirivangani *et al.*, 2021). As far as successful online pharmacy practices are concerned, there are several well-established local and global players existing worldwide like Amazon Pharmacy (USA), NetMeds (India), Lloyds Pharmacy (UK), Clicks Pharmacy (South Africa) and Health Plus Pharmacy (United Arab Emirates).

In the relevant literature, merely a few studies focus their attention on the adoption of online pharmacies among customers in the Indian context. Consequently, there seems a notable occasion to validate the hypothesized model to identify the impact of perceived trust on customer adoption of online pharmacies. In such a situation, an analysis of linking perceived trust with the unified theory of acceptance and use of technology (UTAUT) model becomes vital for efficient online pharmacy policy and regulation. This study expanded the horizon of existing literature on consumer behavior toward online pharmacies by integrating the UTAUT theory (Venkatesh and Davis, 2000), while simultaneously analyzing the moderating influence of perceived trust in the proposed framework.

This analysis is dedicated to elaboration on the impact of the extended UTAUT framework constructs on consumers' adoption intention of online pharmacies. The constructs elected for this study (from UTAUT) are directly relevant to the adoption of online pharmacies as they address important issues for comprehending how different factors influence adoption intentions, with perceived trust, social influences, effort and performance expectancy. In the purview of the technological adaptability of consumers, the analysis of

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

moderating influences of consumers' trust on their adoption knowledge, perception and intention is a unique contribution to the field.

1.1 Overview of online pharmacies

An electronic pharmacy, sometimes referred to as an online pharmacy, is a Web-based digital marketplace where customers can purchase prescription drugs and other medical supplies. In addition to dispensing drugs, a lot of online pharmacies also provide extra services like online consultations with certified pharmacists or other medical professionals, access to health information, and automated prescription refills that help patients take their medications as prescribed (Misra, 2024). Certain platforms also provide telemedicine services, which let patients consult with doctors and get digital prescriptions that the e-pharmacy can fill. Online pharmacies can be two types: first is legitimate, licensed online pharmacies with safe, vetted goods and services that are run under government supervision and legislation. Whereas, second is unregulated or rogue online pharmacies have the potential to sell unsafe or counterfeit drugs and may not require legitimate prescriptions (Bansal *et al.*, 2022).

The e-pharmacies are different from e-commerce as both are engaged with selling of goods in online mode but e-pharmacies and e-commerce platforms differ greatly in terms of how they operate, what laws they must follow and what kinds of goods they sell. The term "e-commerce" deals with online selling of consumer goods, such as electronics and clothes. Whereas, e-pharmacies are a specific subset of e-commerce that sells drugs and other items linked to health care (Sampat and Sabat, 2022).

Examining consumer perception and knowledge as indicators of adoption intention for online pharmacies is the study's aim. Although, knowledge and perception may appear to be separate concepts, but with the right conceptualization, they can still be connected to theories of technology adoption like UTAUT. Despite their differences, these constructs both influence a user's overall intention to use online pharmacies. To bridge this with technology adoption models, there is need to inspect the potential effects of knowledge and perception on behavioral intention, which is the focal variable in models like UTAUT. Subsequently, the behavioral intention to adopt technology is only one part of the adoption process, whereas actual use behavior reflects the final outcome of this intention. That is why analysis is based on intention not behavior.

1.2 Study objectives

The primary objectives of the study are:

- to study the impact of constructs of UTAUT on customer knowledge and perception;
- to investigate the moderating influence of perceived trust on the relationships of constructs of UTAUT and customer knowledge and perception; and
- to determine the relationship between customer knowledge and perception and customer adoption intention.

The structure of this study is categorized into five sections. This section includes introduction (Section 1), literature review and hypothesis (Section 2), methodology (Section 3), data analysis (Section 4) and discussion and conclusion (Section 5).

2. Literature review and hypotheses development

The review of important studies based on prevailing literature is an extensive part of the recognition of study gaps (Kushwaha *et al.*, 2023). Venkatesh and Davis (2000) defined the constructs of effort expectancy, performance expectancy and social influence. These

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

three builds as social influence, effort expectancy and performance expectancy from the UTAUT approach affected consumers' attitudes toward mobile-based internet usage (Wang and Wang, 2010). The market for online pharmacies is anticipated to grow to US$0.92bn by 2024. By 2029, the user penetration rate is predicted to reach 10.04%, from 6.72% in 2024 (Bansal *et al.*, 2022).

2.1 Effort expectancy

Effort expectancy designates the ease of use associated with the technology (Venkatesh and Davis, 2000). A seamless user interface and intuitive navigation are crucial for technology adoption. Effort expectancy alludes to the level of convenience linked with using a technology-based structure. The level of ease of use that a person anticipates from a system is known as effort expectancy (EE). It is a factor that has the power to greatly affect someone's intention to use a system, particularly one that involves technology. The more buyers of a Web-based shopping platform consider it to be a convenient activity and execute less effort to learn innovation, the more the customers have used innovative operations (Venkatesh *et al.*, 2003). Chen and Chang (2013) examined the UTAUT model and found that effort expectancy significantly affects performance expectancy:

H1. Effort expectancy is positively related to consumers' knowledge and perception of online pharmacies.

H2. Effort expectancy is positively associated with consumers' performance expectancy.

H3. Effort expectancy is positively related to social influence.

2.2 Performance expectancy

Performance expectancy refers to the degree to which an individual believes that using a technology will enhance their job performance (Venkatesh *et al.*, 2003). For instance, in e-commerce platforms, performance expectancy relates to the speed, convenience and reliability of transactions. Performance expectancy states how a technique tends to generate advantages for its users while doing the associated activities. Lee and Song (2013) found that performance expectancy and social influence significantly influence consumer behavior. Effort expectancy significantly affects performance expectancy. Performance expectancy and effort expectancy are significant indicators of consumers' attitudes toward mobile-based net banking services during the early phases of adoption (Alsheikh and Bojei, 2014):

H4. Consumers' performance expectancy is positively related to consumers' knowledge and perception of online pharmacies.

2.3 Social influence

Social influence reflects the extent to which consumers perceive that important others (friends, family or colleagues) believe they should use a particular technology (Venkatesh *et al.*, 2003). This construct becomes significant in collectivist cultures like India. The previous research shows that social influence casts a critical influence on consumer attitude, further influencing consumers' buying intentions (Gunawan and Huarng, 2015). An empirical study on the consumer decision-making process and social commerce design suggested that the convenience of feature usage facilitates increased performance for social commerce users. The study indicated that the structure supports consumers' interactions with

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

social commerce, leading to increased utilization of information and services and thus enhanced involvement of customers (Huang and Benyoucef, 2017):

H5. Social influence is positively related to consumers' knowledge and perception of online pharmacies.

2.4 Customer knowledge, perception and adoption intention

Customer knowledge and perception comprises how well consumers understand the technology and form opinions about its value. It includes familiarity, perceived usefulness and potential benefits (Alfahad *et al.*, 2015). The customer knowledge and perception represent customers' awareness and evaluation of products or services offered by marketers (Yin *et al.*, 2016). The adoption intention can be the likelihood of a particular behavior displayed by an individual (Venkatesh *et al.*, 2003). The effort expectancy, performance expectancy and adoption intention of lecturers in Nigeria toward using Google Classroom are positive (Mohammed and Adamu, 2021). Venkatesh and Davis (2000) argued that the users' acceptance is related to identify social influence forms. Thamizhvanan and Xavier (2013) suggested that the consumers' trust significantly influences their adoption behavior. The study also indicated that male buys medicines online more frequently as compared to their female counterparts. The e-commerce research calls for investigating the linkages of trust, privacy protection and data security due to with consumers' buying behavior (Phang *et al.*, 2019). Gupta *et al.* (2018) supported the conceptual and experimental capacity of UTAUT to anticipate the consumers' adoptive behavior. The results depicted that the buyers' selection is essentially associated with performance expectancy, social influence and perceived trust. In addition, technology usage is influenced positively by performance expectancy and social influence. Therefore, it is hypothesized that:

H6. Consumers' knowledge and perception of online pharmacies is positively related to consumers' adoption intention.

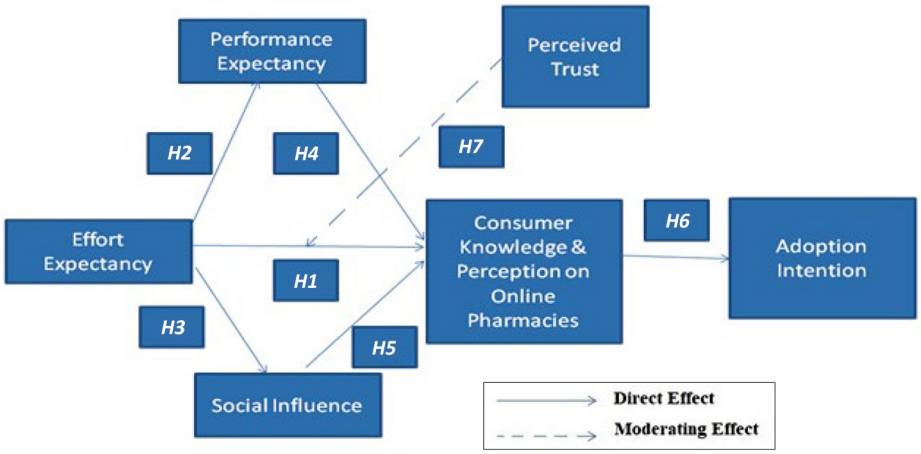
2.5 Perceived trust and antecedents of consumers' knowledge and perception

Perceived trust is a critical factor in mitigating uncertainty in online transactions (Stouthuysen (2020). It encompasses trust in the platform's security, vendor authenticity and data privacy. Perceived trust is the expected behavior of consumers comprising respect they extend toward online buying (Min *et al.*, 2015). Gefen (2000) argued that trust is a basic part of Web-based buying behavior. The simultaneous action of trust, social influence, effort expectancy, performance expectancy and perceived risk represented 55.3% of the variance in consumers' attitude (Lee and Song, 2013). Social factors moderate the relationships between customers' net banking usage and perceived trust, performance expectancy and effort expectancy (Zhang *et al.*, 2018). Zhang *et al.* (2018) suggested that brand trust enhances buyers' adoption intention and faith in a bank that extends net banking facilities to consumers. Furthermore, the interactional need moderated the association between performance expectancy and brand trust. Stouthuysen (2020) argued that generation and building of online trust serve as a mandatory prerequisite to strengthening the effectiveness of e-commerce transactions and inter-organizational associations. Therefore, it is hypothesized that:

H7. Perceived trust moderates the relationship between effort expectancy and consumers' knowledge and perception of online pharmacies.

Based on the articulated hypotheses, Figure 1 shows the hypothesized study model.

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_



Source: Authors' own work

Figure 1. Conceptual study model

2.6 Identified gaps and justification of problem

There are not enough studies in the pertinent literature that focus on how customers adopt online pharmacies in emerging economies like India (Srivastava and Raina, 2021; Sampat and Sabat, 2022). Whereas, limited studies focuses on customer purchase intention and behavior toward online pharmacies (Bansal *et al.*, 2022). Nonetheless, no prior studies focuses its attention on moderating role perceived trust in adoption of online pharmacies (Assin *et al.*, 2024). The abovementioned gaps highlights the lack of comprehensive studies on customer purchase intention and perceived trust in online pharmacies and emphasizes the need for its contextual validation in the Indian setting. There is a dearth of empirical research examining how customer purchase intention influence online pharmacies adoption constructs in the Indian market, where consumer behavior significantly differs. Trust is a major component influencing user behavior in the adoption of online pharmacies. Therefore, users' comfort level with the service is directly impacted by their perception of trust (Misra, 2024).

Perceived trust, however, may moderate the relationships between behavioral intention and other adoption factors (i.e. effort expectancy, performance expectancy and social influence) in addition to having a direct impact on adoption. This is because users may still hesitate if they do not trust a platform, even if they believe it to be beneficial (performance expectancy) or easy to use (effort expectancy). The effects of these factors can be strengthened or weakened by trust. Although, apart from perceived trust, user trust is also considerable but perceived trust choose as a moderator in present study as users faces several challenges in trusting the online pharmacy adoption (Al Halbusi *et al.*, 2024).

The Indian context was selected for this study because it has one of the world's fastest-growing emerging digital economies. It is a great place to study technology adoption because of the country's fast increasing internet penetration, smartphone usage and online services, particularly in the health-care industry, which includes online pharmacies. Subsequently, it is a perfect place to study online pharmacy adoption using UTAUT because of its distinct blend of a developing digital economy, government-led digitization

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

initiatives and health-care issues. This analysis investigated the effect of effort expectancy, performance expectancy and social influence constructs of the UTAUT framework on customer knowledge and perception. The study is also unique in evaluating the moderating influence of customers' perceived trust in the association of effort expectancy and customer knowledge and perception.

3. Research methodology

This portion is dedicated to the methodological part of the analysis as it comprises the explanation of tools and techniques used in this study in the direction to attain the study objectives.

3.1 Sample size and data collection

The study involved participants from the Delhi-National capital region (NCR) of India in evaluating consumer adoption intention toward online pharmacies in India. The reason behind Delhi as residents of all the states of India and thus rightly represent the Indian population. The researcher received 470 responses from the distributed 900 survey instruments. This data collection shows that 52.22% of the respondents approached responded to the survey study, representing an appropriate response rate for a survey-based study. The researchers found a sample size of 451 after screening received responses, which also justifies the adequacy of the sample size. In the case of structural equation modeling (SEM) studies, the respondent size of approximately 200 seems to be appropriate for average SEM models. Also, researchers use Harman's single-factor test to assess the common technique bias, which shows that the responses were free of common method bias as a single-factor solution found below 40% of the threshold value. Thus, the essential assumption of common method bias for conducting CFA and SEM is satisfied. Table 1 displays the demographic details of the respondents of this analysis.

Table 1. Demographic characteristics of the respondents (n = 451)

Characteristic	Category	Frequency	%
Gender	Male	277.00	61.4
	Female	174.00	38.6
Age	15–30 years	265.00	58.8
	30–45 years	157.00	34.8
	45–60 years	28.00	6.2
	60 years above	1.00	0.2
Education	12th	8.00	1.8
	Graduate	189.00	41.9
	PG	171.00	37.9
	Doctorate	73.00	0.16.2
	Other	10.00	2.2
Experience (years)	0–5	100.00	43.6
	6–10	74.00	32.4
	11–15	30.00	13
	16–20	15.00	6.5
	21–25	5.00	2.2
	26–30	2.00	0.8
	30 and above	3.00	1.2

Source: Authors' own work

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

3.2 Questionnaire instruments and sampling strategy

The investigators approach respondents through online social media platforms like LinkedIn, WhatsApp, Facebook, e-mail groups and the researcher's network by using nonprobability-based convenience sampling. The reason behind the selection of convenience sampling as it helps in gathering of data from a large and sufficient number of respondents in a shorter time (Singh and Kushwaha, 2024). Convenience sampling was used due to its practicality and relevance in an exploratory study context of study. While the method has limitations in terms of representativeness, efforts were made to ensure demographic diversity and findings were cross-validated with existing literature.

A cover letter and the survey instrument with author information were shared with the respondents who consented to take part in the study. When necessary, the researcher used video conferencing to help the respondents understand the meaning of the survey items to prevent participants from giving ambiguous answers. Online response collection became the most popular, secure and practical method of gathering data during the COVID-19 pandemic. The constructs chosen for the analysis (e.g. performance expectancy, effort expectancy, social influence, customer knowledge and perception and perceived trust) are derived from well-established frameworks, i.e. UTAUT. These constructs are highly significant for studying adoption of online pharmacies. The survey instruments used for collecting data are.

3.2.1 Effort expectancy. The effort expectancy was evaluated using four-items self-administered scales ranked on a five-point Likert scale (Venkatesh *et al.*, 2003) from "Strongly Disagree" to "Strongly Agree". The Cronbach's alpha of the scale was 0.83.

3.2.2 Performance expectancy. The performance expectancy was assessed using a four-item scale suggested by Venkatesh *et al.* (2003) and Leelayouthayotin (2004) rated on a five-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree". The Cronbach's alpha of the scale was 0.73.

3.2.3 Social influence. The social influence was evaluated using a three-item scale, rated on a five-point Likert scale (Venkatesh *et al.*, 2003) from "Strongly Disagree" to "Strongly Agree". The Cronbach's alpha of the scale was 0.83.

3.2.4 Customer knowledge and perception. The researchers evaluate the customer knowledge and perception using a five-item scale (Alfahad *et al.*, 2015), rated on a seven-point Likert scale from "Not At All" to "Frequently, if Not Always". The Cronbach's alpha of the scale was 0.79.

3.2.5 Perceived trust. The researchers measured the perceived trust using four-item scale with items adapted from Featherman and Pavlou (2003) and Koufaris and Hampton-Sosa (2004). The scale was rated on a five-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree". The Cronbach's alpha of the scale was 0.87.

3.2.6 Adoption intention. The three-item scale (Venkatesh *et al.*, 2003), rated on a five-point Likert scale from "Absolutely True" to "Absolutely Untrue", was used to measure the adoption intention. The Cronbach's alpha of the scale was 0.92.

4. Data analysis

This section used the AMOS 25.0 software to analyze the associations between the constructs, and confirmatory factor analysis (CFA) was used to evaluate the measurement model. Also, researchers further deployed SEM to test the predetermined hypotheses. This analysis assesses the assumptions using significance levels of relationships among the study constructs found from SEM analysis by bootstrapping the samples 5,000 times, indicating the preciseness of the analytical strategy. The model fit was assessed using RMSEA (0.07), CFI (0.92) and SRMR (0.06), all of which fall within the acceptable

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

thresholds (RMSEA < 0.08, CFI > 0.90, SRMR < 0.08). These metrics were chosen for their robustness in evaluating SEM models, balancing parsimony and predictive accuracy.

4.1 Analysis of measurement model

The findings related to correlations, reliability, descriptive statistics and validity are presented in Table 2. The Cronbach's alpha (Cronbach, 1951) and composite reliability of the scales emerge to be more than 0.70 indicating the appropriate reliability of the scales. Also, the average variance extracted of all the subscales evolved to be more than 0.50 (Fornell and Larcker, 1981) and factor loadings come out to be significant at 0.0001 level (Table 3), indicating the appropriate validity of the scales.

The CFA analysis conducted by the researcher depicted that the data was found to be having good fit with the proposed model. The measurement model shows six factor solution as: X^2 [204] = 488.002, CFI = 0.95, IFI = 0.95, TLI = 0.94, RMSEA = 0.056, SRMR = 0.04 and P Close = 0.000 (Hu and Bentler, 1999). The SEM results for the structural model (Figure 2) show a good fit with the data as; X^2 [212] = 557.697, $p < 0.01$; CFI = 0.94; TLI = 0.93; IFI = 0.94; RMSEA = 0.06; SRMR = 0.05.

4.2 Analysis of structural modal and hypotheses testing

The SEM analysis results revealed that *H1* is accepted. The effort expectancy was found to be positively influencing customer knowledge and perception (E = 0.63, $p < 0.01$) as shown in Table 4. Similarly, *H2–H6* are accepted as effort expectancy was also found to be positively influencing performance expectancy (E = 0.78, $p < 0.01$) and social influence (E = 0.66, $p < 0.01$), performance expectancy (E = 0.49, $p < 0.01$) and social influence (E = 0.14, $p < 0.05$) were found to be positively influencing customer knowledge and perception, and customer knowledge and perception was found to be positively influencing adoption intention (E = 0.90, $p < 0.01$) as shown in Table 4. Figure 3 represents the hypothesized structural model.

Perceived trust satisfied the criteria of being a moderator by meeting the three criteria suggested by Baron and Kenny (1986) as perceived trust is not directly related to customer knowledge and perception. Second, the interaction (effort expectancy X perceived trust) emerged to be significant, and finally, it acted as an independent variable. Furthermore, moderated structural equation modeling (MSEM) suggested by Ping (1995) was used to evaluate the moderating influence of perceived trust on the relationship between effort expectancy and customer knowledge and perception. The model shows a good fit with the data as; X^2 = 2.4.666, $p < 0.01$; CFI = 0.95; TLI = 0.94; IFI = 0.95; RMSEA = 0.06; SRMR = 0.04. Also, MSEM analysis supported the proposed hypothesis *H7* as the perceived trust emerged to be moderating the relationship of effort expectancy and customer knowledge and perception (interaction effect on consumer knowledge and perception = 0.024, standard error = 0.009, p = 0.007). Figure 2 shows that the moderating influence of perceived trust strengthens the positive relationship between effort expectancy and knowledge and perception.

5. Discussion and conclusive remarks

This study conceptually and empirically indicated that the consumers' performance expectancy and effort expectancy served as crucial determinants of consumers' adoption intention, knowledge and perception of online pharmacies. This study systematically and logically validated the moderating role of perceived trust in influencing effort expectancy toward customers' perception of online pharmacies in the Indian market. Furthermore, the study indicated that social influence too casts a significant impression on consumers' adaptive tendencies toward online pharmacies. Finally, the study enlightens the research

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

Table 2. Descriptive statistics, reliability, AVE and correlations

S. no.	Construct	X	μ	1	2	3	4	5	6	7	8	9	α	β
1	Gender	1.61	0.48											
2	Age	2.47	0.62	0.28**										
3	Education	2.75	0.82	0.12**	0.50**									
4	Effort expectancy	3.10	0.84	−0.07	0.06	0.08	*0.56*						0.83	0.75
5	Performance expectancy	3.88	0.79	−0.03	−0.01	−0.03	0.39**	*0.54*					0.73	0.85
6	Social influence	3.48	0.93	0.01	0.02	0.06	0.33**	0.31**	*0.54*				0.83	0.74
7	Customer knowledge and perception	3.56	0.87	−0.02	0.04	0.09	0.32**	0.29**	0.63**	*0.52*			0.79	0.86
8	Perceived trust	2.77	0.97	0.07	−0.03	0.01	0.14**	0.21**	0.30**	0.35**	*0.53*		0.88	0.81
9	Adoption intention	3.41	0.62	−0.05	0.01	0.10*	0.23**	0.11*	0.33**	0.36**	0.05*	*0.54*	0.93	0.82

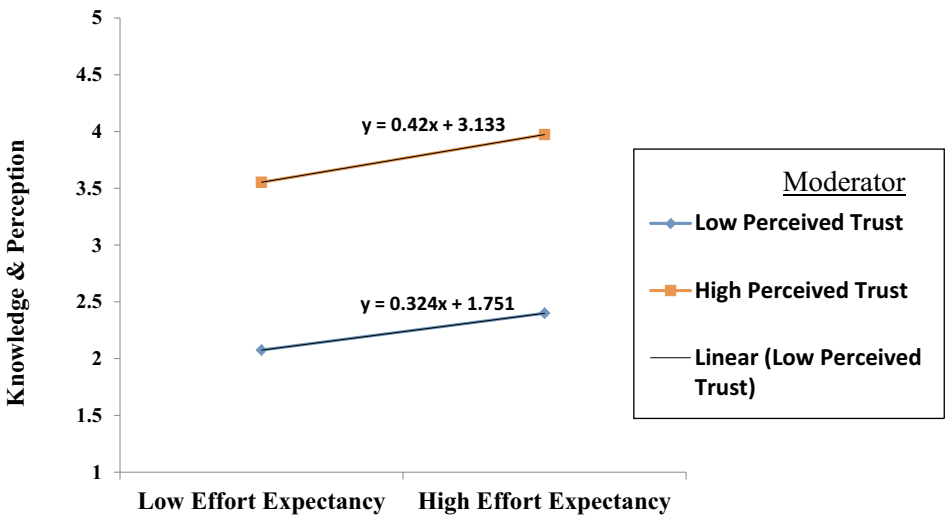
Notes: **$p < 01$; *$p < 05$; X = mean; μ = standard deviation; α = Cronbach's alpha; β = composite reliability; average variance extracted (AVE) are mentioned across parentheses diagonally
Source: Authors' own work

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/1061414 2/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW

Table 3. Factor extraction and study constructs model fit evaluation

Measure	Measure item	Factor loading
Effort expectancy	EE1	0.82
	EE2	0.80
	EE3	0.83
	EE4	0.70
Performance expectancy	PE1	0.84
	PE2	0.84
	PE3	0.60
	PE4.	0.52
Social influence	SI1	0.81
	SI2	0.82
	SI3	0.74
Perceived trust	PT1	0.83
	PT2	0.80
	PT3	0.83
	PT4	0.70
Customer knowledge and perception	KP1	0.88
	KP2	0.87
	KP3	0.83
	KP4	0.82
	KP5	0.80
Adoption intention	AI1	0.92
	AI2	0.86
	AI3	0.93

Source: Authors' own work



Source: Authors' own work

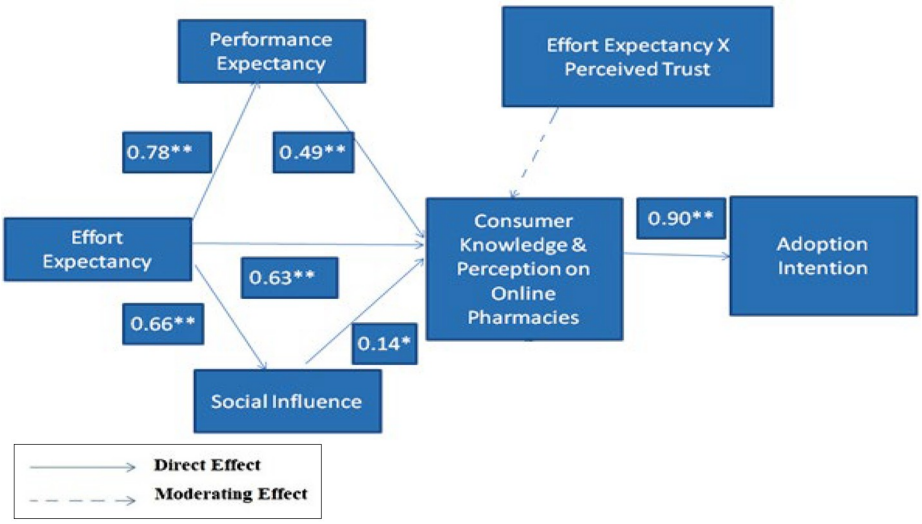
Figure 2. Moderating influence of perceived trust

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

Table 4. SEM analysis results

Relationship among the constructs	Standardized direct effects	Decision
H1: Effort expectancy → Knowledge and perception	0.637**	Supported
H2: Effort expectancy → Performance expectancy	0.787**	Supported
H3: Effort expectancy → Social influence	0.667**	Supported
H4: Performance expectancy → Knowledge and perception	0.497**	Supported
H5: Social influence → Knowledge and perception	0.141*	Supported
H6: Knowledge and perception → Adoption intention	0.906**	Supported

Notes: NE = no effect; *$p < 0.05$; **$p < 0.01$; $n = 451$
Source: Authors' own work



Source: Authors' own work

Figure 3. Hypothesized structural model

inventory of online pharmacies by exploring several determinants of consumers' knowledge and perception of online pharmacies. Despite the growing adoption of online pharmacies, significant challenges persist, particularly in understanding the role of consumer trust, security concerns, digital divide, a lack of physical interaction, regulatory challenges and logistics delays determining online pharmacies adoption behavior.

The online pharmacy industry is gaining rapid momentum in India through ongoing technological and infrastructural developments and the positively adaptive response from Indian consumers. The COVID-19 pandemic has caused people to think beyond the physical purchase of medicines and adopt digital transactions that are entirely safe and convenient. Also, the technological wonders such as smartphone applications and other self-service technologies are further generating high-speed diffusion of online interactions of consumers toward products and services usage and consumption (Gupta *et al.*, 2018). The results and findings of the present study will surely assist researchers in

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

further exploring the untapped determinants of the consumers' online perception and knowledge of online pharmacies in future studies.

The findings on study align with UTAUT-based studies in similar developing contexts, e.g. Nigeria and South Africa. Although, the findings are similar to the previous studies and assumptions of UTAUT theory for adoption of online pharmacies remains relevant (Sampat and Sabat, 2022; Al Halbusi *et al.*, 2024). This study validated an innovative understanding of associations between the constructs of effort expectancy, performance expectancy, social influence, consumers' knowledge and perception of online pharmacies. The present study recommended that the consumers' effort expectancy directly influences consumers' ability and perception toward online pharmacies and indirectly through performance expectancy and social influence.

The online pharmaceutical industry should develop the online buying process as per the attributes of Indian society. The online process will make the buying procedures convenient, ensuring transaction security, fast delivery to consumers and enhanced commitment to providing original medicines. These recommendations will boost the performance expectancy and the effort expectancy of buying drugs through online pharmacies. Subsequently, a supplementary package of online consultation instructions or facilitation to the use of prescribed medicines may enhance trust among online buyers. This study revealed several outcomes regarding the preferences of people toward online pharmacies in India. Indeed, the impact of the pandemic has caused further support to adopt safety measures for buying medicines for consumers. The technological advancements and growing population of e-commerce adopters further improve the opportunities for online pharmacies. At last, there exists enormous potential for the growth and expansion of online pharmacies in India and this seems to be just the beginning of the same.

5.1 Theoretical implications

The consequences of this analysis are twofold. First, it contributes to comprehensive theory building of online pharmacy literature. Second, it acknowledges the elements of online pharmacy adoption among customers that can be used to enhance its adoption by pharmacy practitioners. The outcomes of this analysis contributed to the practical evidence by detecting the role of perceived trust along with the elements of the UTAUT model. The present study extended the UTAUT model in the Indian online pharmacy industry by suggesting the unique moderating role of perceived trust in the association of effort expectancy and knowledge and perception. The finding enriches the literature on the linkage of effort expectancy, performance expectancy and consumers' knowledge and perception toward online pharmacies additionally to suggestions extended in UTAUT model. The findings confirms that adoption of online pharmacies is significantly influenced by perceived trust in UTAUT and offers a more holistic understanding of consumer behavior in online health-care settings.

5.2 Practical implications

Online pharmacies can help governments ensure broader access to medications, especially for populations in remote areas. Governments can potentially reduce the burden on public health-care infrastructure by minimizing the need for in-person visits for prescription refills. Online pharmacies can support national health programs by integrating digital health services, promoting preventive care and improving health-care access. Online pharmacies can also help pharmacists reach a wider audience by extending their pharmaceutical services, like drug therapy consultations, chronic disease management and medication counseling. Due to effective online pharmacy services, pharmacies can now reach consumers directly and gain a greater market share by reducing their reliance on middlemen, especially in places

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

where physical pharmacies might not be easily accessible. Online pharmacies can help hospitals and clinics streamline prescription management, improve patient care continuity, reduce patient load for minor prescriptions and access comprehensive medication data. In addition, patients can benefit from online pharmacies' round-the-clock accessibility, ease, privacy protection and lower overhead costs, particularly for older or disabled patients.

5.3 Limitations and further research

This analysis possesses some restrictions like any study (Singh *et al.*, 2021). In this analysis, merely particular application constructs have been selected from the UTAUT model as per the need of the study. The survey sample of this analysis is restricted only to the online pharmacy consumers of the Delhi NCR region. The future analysis may perform a study comprising consumers from other metropolitan cities of India. Further researchers may use this framework as a foundation to further examine the impact of various technological transformations on the usage of e-pharmacy consumers. It is recommended for future scholars to use probability sampling techniques to enhance external validity.

References

Al Halbusi, H., Al-Sulaiti, K., Abdelfattah, F., Ahmad, A.B. and Hassan, S. (2024), "Understanding consumers' adoption of e-pharmacy in Qatar: applying the unified theory of acceptance and use of technology", *Journal of Science and Technology Policy Management*.

Alfahad, N.F., Albelali, M.T., Khurshid, F., Al-Arifi, M.N., Al-Dhawailie, A.A. and Alsultan, M.S. (2015), "Perception and knowledge to online pharmacy services among consumers in Riyadh, Saudi Arabia: a pilot survey", *Lat Am J Pharm*, Vol. 34 No. 6, pp. 1113-1118.

Alsheikh, L. and Bojei, J. (2014), "Determinants affecting customer's intention to adopt mobile banking in Saudi Arabia", *Int. Arab. J. e Technol*, Vol. 3 No. 4, pp. 210-219.

Altinay, L. and Poudel, S. (Eds) (2015), *Enhancing Customer Experience in the Service Industry: A Global Perspective*, Cambridge Scholars Publishing, Newcastle upon Tyne.

Assin, T.J., George, N., Aboobaker, N., P, S. and V., A. (2024), "Emerging market dynamics: risk perceptions, perceived usefulness and E-pharmacy adoption", *International Journal of Pharmaceutical and Healthcare Marketing*, doi: 10.1108/IJPHM-11-2023-0101.

Bansal, S., Kaur, H., Mahendiratta, S., Sarma, P., Kumar, S., Sharma, A.R., Joshi, R., Bhattacharyya, A., Prajapat, M., Prakash, A. and Medhi, B. (2022), "A preliminary study to evaluate the behavior of Indian population toward E-pharmacy", *Indian Journal of Pharmacology*, Vol. 54 No. 2, pp. 131-137.

Baron, R.M. and Kenny, D.A. (1986), "The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations", *Journal of Personality and Social Psychology*, Vol. 51 No. 6, p. 1173.

Chen, K.Y. and Chang, M.L. (2013), "User acceptance of 'near field communication' mobile phone service: an investigation based on the 'unified theory of acceptance and use of technology' model", *The Service Industries Journal*, Vol. 33 No. 6, pp. 609-623.

Connell, J. (2008), "Towards a global health care system?" *The International Migration of Health Workers*, Routledge, New York, pp. 17-45.

Cronbach, L.J. (1951), "Coefficient alpha and the internal structure of tests", *Psychometrika*, Vol. 16 No. 3, pp. 297-334.

Dcruz, A.C., Mokashi, V.N., Pai, S.R. and Sreedhar, D. (2022), "The rise of E-pharmacy in India: benefits, challenges, and the road ahead", *Indian Journal of Pharmacology*, Vol. 54 No. 4, pp. 282-291.

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

Dubey, A. and Verma, A.S. (2022), "Effective remote healthcare and telemedicine approaches for improving digital healthcare systems", *Digital Health Transformation with Blockchain and Artificial Intelligence*, CRC Press, New York, pp. 273-297.

Elbeddini, A. and Yeats, A. (2020), "Pharmacist intervention amid the coronavirus disease 2019 (COVID-19) pandemic: from direct patient care to telemedicine", *Journal of Pharmaceutical Policy and Practice*, Vol. 13 No. 1, p. 23.

Featherman, M.S. and Pavlou, P.A. (2003), "Predicting e-services adoption: a perceived risk facets perspective", *International Journal of Human-Computer Studies*, Vol. 59 No. 4, pp. 451-474.

Fornell, C. and Larcker, D.F. (1981), "Structural equation models with unobservable variables and measurement error: Algebra and statistics", *Journal of Marketing Research*, Vol. 18 No. 3, pp. 382-388.

Gefen, D. (2000), "E-commerce: the role of familiarity and trust", *Omega*, Vol. 28 No. 6, pp. 725-737.

Gunawan, D.D. and Huarng, K.H. (2015), "Viral effects of social network and media on consumers' purchase intention", *Journal of Business Research*, Vol. 68 No. 11, pp. 2237-2241.

Gupta, A., Dogra, N. and George, B. (2018), "What determines tourist adoption of smartphone apps? An analysis based on the UTAUT-2 framework", *Journal of Hospitality and Tourism Technology*, Vol. 9 No. 1, pp. 50-64.

Hu, L.T. and Bentler, P.M. (1999), "Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives", *Structural Equation Modeling: A Multidisciplinary Journal*, Vol. 6 No. 1, pp. 1-55.

Huang, Z. and Benyoucef, M. (2017), "The effects of social commerce design on consumer purchase decision-making: an empirical study", *Electronic Commerce Research and Applications*, Vol. 25, pp. 40-58.

Koufaris, M. and Hampton-Sosa, W. (2004), "The development of initial trust in an online company by new customers", *Information and Management*, Vol. 41 No. 3, pp. 377-397.

Kushwaha, J., Sharma, A. and Singh, P. (2023), "Exploration and prioritization of enablers to organization work–family balance planning for working sole Indian mothers integrating fuzzy Delphi and AHP", *International Journal of Social Economics*, Vol. 50 No. 3, pp. 398-418.

Lee, J.H. and Song, C.H. (2013), "Effects of trust and perceived risk on user acceptance of a new technology service", *Social Behavior and Personality: An International Journal*, Vol. 41 No. 4, pp. 587-597.

Leelayouthayotin, L. (2004), "Factors influencing online purchase intention: the case of health food consumers in Thailand" (Doctoral dissertation, University of Southern Queensland).

Mendoza, R.L. (2014), "Internet medicine and pharmacies: private choice, public risks, and healthcare business", *International Journal of Healthcare Management*, Vol. 7 No. 3, pp. 185-199.

Min, H., Kim, H.J. and Lee, S.B. (2015), "Extending the challenge–hindrance stressor framework: the role of psychological capital", *International Journal of Hospitality Management*, Vol. 50, pp. 105-114.

Misra, P. (2024), "Purchase intention toward E-pharmacy: the consumption value perspective", *International Journal of Pharmaceutical and Healthcare Marketing*, doi: 10.1108/IJPHM-12-2023-0107.

Mohammed, F.N. and Adamu, Z.E. (2021), "Investigation into performance expectancy, effort expectancy and acceptances to use google classroom for instruction among college of education lectures in Niger state", *IJER-International Journal of Educational Research*, Vol. 4 No. 2, pp. 114-124.

Phang, D.C., Wang, K., Wang, Q., Kauffman, R.J. and Naldi, M. (2019), "How to derive causal insights for digital commerce in China? A research commentary on computational social science methods", *Electronic Commerce Research and Applications*, Vol. 35, p. 100837.

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

Ping, R.A. Jr, (1995), "A parsimonious estimating technique for interaction and quadratic latent variables", *Journal of Marketing Research*, Vol. 32 No. 3, pp. 336-347.

Sampat, B. and Sabat, K.C. (2022), "Adoption of online pharmacies in India: an empirical study", *International Journal of Business Innovation and Research*, Vol. 29 No. 4, pp. 449-478.

Singh, P. and Kushwaha, R. (2024), "Envisaging Indian farmers' desires from agricultural index insurance integrating rank sum weighting method and MoSCoW technique: an approach to requirements prioritization", *International Journal of Social Economics*, doi: 10.1108/IJSE-03-2024-0233.

Singh, P., Kushwaha, R. and Kushwaha, J. (2021), "Assessing visitors satisfaction about museum service quality using Kano model", *2021 IEEE 9th Region 10 Humanitarian Technology Conference (R10-HTC), IEEE*, pp. 1-6.

Srivastava, M. and Raina, M. (2021), "Consumers' usage and adoption of e-pharmacy in India", *International Journal of Pharmaceutical and Healthcare Marketing*, Vol. 15 No. 2, pp. 235-250.

Stouthuysen, K. (2020), "A 2020 perspective on "the building of online trust in e-business relationships", *Electronic Commerce Research and Applications*, Vol. 40, p. 100929.

Tan, B.Y., Chew, N.W., Lee, G.K., Jing, M., Goh, Y., Yeo, L.L., Zhang, K., Chin, H.K., Ahmad, A., Khan, F.A. and Shanmugam, G.N. (2020), "Psychological impact of the COVID-19 pandemic on health care workers in Singapore", *Annals of Internal Medicine*, Vol. 173 No. 4, pp. 317-320.

Teodorescu, C.A., Durnoi, A.N.C. and Vargas, V.M. (2023), "The rise of the mobile internet: tracing the evolution of portable devices", In *Proceedings of the International Conference on Business Excellence*, Vol. 17 No. 1, pp. 1645-1654.

Thamizhvanan, A. and Xavier, M.J. (2013), "Determinants of customers' online purchase intention: an empirical study in India", *Journal of Indian Business Research*, Vol. 5 No. 1, pp. 17-32.

Tirivangani, T., Alpo, B., Kibuule, D., Gaeseb, J. and Adenuga, B.A. (2021), "Impact of COVID-19 pandemic on pharmaceutical systems and supply chain–a phenomenological study", *Exploratory Research in Clinical and Social Pharmacy*, Vol. 2, p. 100037.

Venkatesh, V. and Davis, F.D. (2000), "A theoretical extension of the technology acceptance model: four longitudinal field studies", *Management Science*, Vol. 46 No. 2, pp. 186-204.

Venkatesh, V., Morris, M.G., Davis, G.B. and Davis, F.D. (2003), "User acceptance of information technology: toward a unified view", *MIS Quarterly*, Vol. 27 No. 3, pp. 425-478.

Wang, H.Y. and Wang, S.H. (2010), "User acceptance of mobile internet based on the unified theory of acceptance and use of technology: investigating the determinants and gender differences", *Social Behavior and Personality: An International Journal*, Vol. 38 No. 3, pp. 415-426.

Wen, H.J. and Tan, J. (2003), "The evolving face of telemedicine and e-health: opening doors and closing gaps in e-health services opportunities and challenges", *36th Annual HI International Conference on System Sciences, 2003. Proceedings of the, IEEE*, p. 12.

Wirtz, B.W., Schilke, O. and Ullrich, S. (2010), "Strategic development of business models: implications of the Web 2.0 for creating value on the internet", *Long Range Planning*, Vol. 43 Nos 2/3, pp. 272-290.

Worlu, R., Kehinde, O.J. and Borishade, T.T. (2016), "Effective customer experience management in health-care sector of Nigeria: a conceptual model", *International Journal of Pharmaceutical and Healthcare Marketing*, Vol. 10 No. 4, pp. 449-466.

Yin, M., Li, Q. and Qiao, Z. (2016), "A study on consumer acceptance of online pharmacies in China", In *Proceedings of the 18th Annual International Conference on Electronic Commerce: e-Commerce in Smart connected World*, pp. 1-8.

Zhang, Y., Weng, Q. and Zhu, N. (2018), "The relationships between electronic banking adoption and its antecedents: a meta-analytic study of the role of national culture", *International Journal of Information Management*, Vol. 40, pp. 76-87.

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_

About the authors

Sanjeev Saraswat is an Assistant Professor of Marketing domain in Institute of Business Management at GLA University, Mathura, India. He completed his doctoral research in marketing domain at Jiwaji University, Gwalior, Madhya Pradesh. He has more than 20 years of experience in academics.

Pankaj Singh is an Associate Professor of Marketing Science in School of Business at Galgotias University, Greater Noida, India. He completed his doctoral research in marketing domain at ABV-Indian Institute of Information Technology and Management, Gwalior, Madhya Pradesh. He holds a masters' in International Business Management and M. Phil. in Marketing and his particular research interest in the area of Service Marketing, Marketing, Social Media and Public Policy. He published several journal articles that indexed in Scopus and WOS-ESCI, listed in ABDC: B-C, ABS: 1-2 rank. He has also published case-studies in ABS-1 listed journals and several IEEE conference proceedings and book chapters in all leading publishers i.e., Springer, IGI Global, CRC Press, Wiley and Emerald publications. He has more than 14 years of experience in academics and industry. Pankaj Singh is the corresponding author and can be contacted at: pankajabviiitm@gmail.com

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**

Downloaded from http://www.emerald.com/ijphm/article-pdf/19/4/964/10614142/ijphm-05-2024-0042en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_